November 22, 2025

Webb Motorsports UDDS Regenerative Braking Analysis

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Major Deliverables

1. Custom numerical code for high-accuracy integration and differentiation result in the following baseline values:
   a) **Total Regenerative Energy: 0.06424 kWh**
   b) **Total Distance Traveled: 11.991 km**

2. Noise analysis on the same base dataset to include Gaussian jitter, dropout, quantization effects resulted in:
   a) **Design your sensor package for MAXIMUM allowable sensor jitter: 0.145 m/s**

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In our analysis, we created a custom high-accuracy differentiation and integration (betterDifferentiator, betterIntegrator) to process UDDS data as a benchmark given your parameters, resulting in:
Total Regenerative Energy: 0.06424 kWh
Total Distance Traveled: 11.991 km
We confirmed the baseline drivetrain performance under ideal measurement conditions.

For our noise analysis for your sensor package sensor sensitivity, we included Gaussian jitter, dropout, and quantization effects and applied those to the baseline dataset, comparing error to give the:
Maximum allowable sensor jitter: 0.145 m/s
This rovides quantitative limits for sensor noise, and at this standard will ensure accurate regenerative braking estimates.

Key Takeaway: Sensors must meet or exceed this jitter threshold to maintain reliable distance and energy calculations.

Outline

1. Develop Improved Numerical Methods & Compute Baseline Regenerative Energy
   - Methods and Assumptions – Numeric Code
   - Methods and Assumptions – Regenerative Data
   - Results
   - Analysis and Conclusions

2. Analyze Noise Sensitivity and Determine Allowable Jitter
   - Methods and Assumptions
   - Results
   - Analysis and Conclusions

3. Conclusions and Recommendations

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Develop Improved Numerical Code

Use Improved methods to determine baseline Regenerative Data

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Methods and Assumptions – Numeric Code

MATLAB's base numeric integrator trapz() does not provide the best high-level accuracy needed to integrate later shown Power equation to give Energy.

- betterIntegrator() adaptively chooses best higher level integration rule per interval
  - Simpson's 1/3 Rule: global $O(h^4)$ error, requires 3 evenly spaced points
  - Simpson's 3/8 Rule: global $O(h^4)$ error, requires 4 evenly spaced points
  - Trapezoidal rule: global $O(h^2)$ error

- **Simpson's Rules are only valid on uniformly spaced intervals**
  - Function falls back to Trapezoidal rule for uneven intervals

- Further assumes data smooth enough for polynomial approximation

Function	betterIntegrator() Error %	Trapz() Error %
f(x) = x^2	1.6e-14	.5
f(x) = sin(x)	5.56e-5	.083
f(x) = exp(-x)	5.54e-5	.083

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MATLAB's trapz() is only second-order accurate. With long, noisy driving data or in applications that require high level accuracy, this method will result in significant cumulative error.

Simpson's 1/3 and 3/8 rules integrate a function by curve fitting low-order polynomials over small windows:
- 1/3 rule: quadratic fit over 3 evenly spaced points
- 3/8 rule: cubic fit over 4 evenly spaced points

Both give fourth-order accuracy O(h^4), significantly better than trapezoidal.

Simpson's methods are only valid when the underlying x-spacing is uniform. Real sensor data often violates that, especially after resampling or missing packets.

The custom function checks the spacing at every step and adaptively applies the most accurate valid rule:
Use Simpson 1/3 if 3 uniform points exist
Use Simpson 3/8 at the end if 4 uniform points exist leftover at the end
Otherwise fall back to trapezoidal

This guarantees the highest-order rule is used whenever allowed by the dataset.

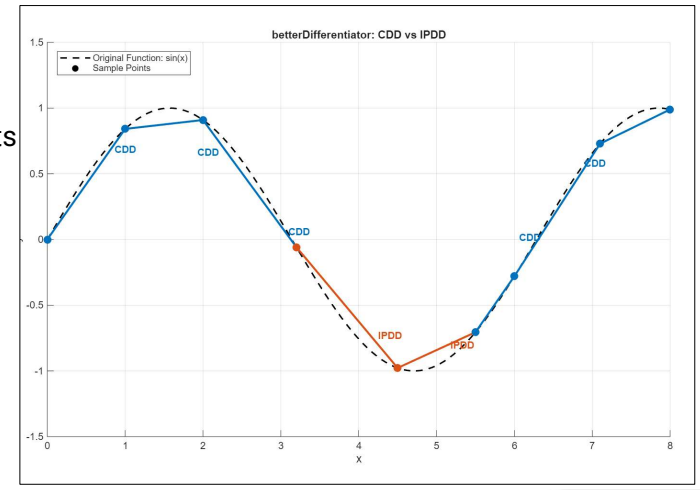
Assumptions:
- Local spacing must be close to uniform for Simpson's rules
= Signals must be smooth enough for low-order polynomial fits (vehicle velocity and power commonly are - this is a valid method for your purposes).

The shown table gives a first level demonstration on the higher accuracy of the betterIntegration() function with error related to the actual value vs trapz() error. Note that the main significance of betterIntegration() is the interval spacing adaptability to provide highest accuracy possible for a dataset.

Methods and Assumptions – Numeric Code

Numeric differentiation is highly susceptible to error. Differentiation is required in Regenerative energy calculation. A custom $O(h^2)$ accuracy method is required.

- betterDifferentiator() adaptively selects the most relevent form per interval
  - Central Divided Differences: $O(h^2)$ error for uniform data
  - Interpolating Polynomial Divided Differences: $O(h^2)$ error for nonuniform data
- Assumes:
  - Local smoothness
  - Spacing irregularities are small
  - Moderate noise levels



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Numeric differentiation amplifies noise and is highly susceptible to error. In regenerative energy calculations, accurate derivatives of velocity are critical. Standard forward or central differences on unevenly spaced or noisy data can accumulate significant errors.

The custom function betterDifferentiator() adaptively chooses the most accurate method for each interval:

Central Divided Differences (CDD): O(h^2) error, used when three consecutive points are approximately uniform. Using 3 points in the form gives the associated O(h^2) error

Interpolating Polynomial Divided Differences (IPDD): O(h&2) error, used when spacing is irregular, fitting a second-order polynomial locally.

The first and last points are handled with forward/backward schemes due to future and past points needed for CDD. This mathematical form is similar to CDD.

The method, as most differentiators necessitate, assume:

-Local signal smoothness sufficient for polynomial approximation
-Spacing irregularities between points are moderate
-Noise levels are not extreme

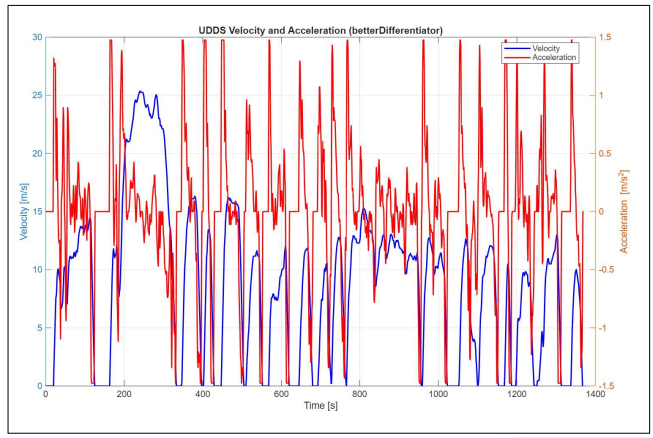
The plot shows the adaptive nature of the plot, and how if given uneven data, the method will switch, providing O(h^2) error convergence universally.

The main significance of betterDifferentiator() is that it maintains requested O(h^2) accuracy across both uniform and nonuniform intervals, ensuring reliable derivative estimates for the following energy calculations.

Methods and Assumptions – Regen Dat

- To evaluate the given P_regen equation, acceleration is needed – derivative of UDDS velocity.
- Integration done on P_regen for Energy.

$$P_{\text{regen}}(t) = \eta \cdot \min(P_{\text{clip}}, \max(0, -m \cdot a(t) \cdot v(t)))$$



- Vehicle Parameters: Mass = 1500 kg, Regen efficiency = 0.75, Max regen power = 80 kW
- Dataset: UDDS velocity Data
- Assumptions - UDDDS data has:
  - local spacing uniform
  - Dropout and noise negligible in baseline
  - Sufficiently smooth signals for numerical methods

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The goal of this analysis is to quantify the vehicle's regenerative braking potential over a standard UDDS drive cycle.

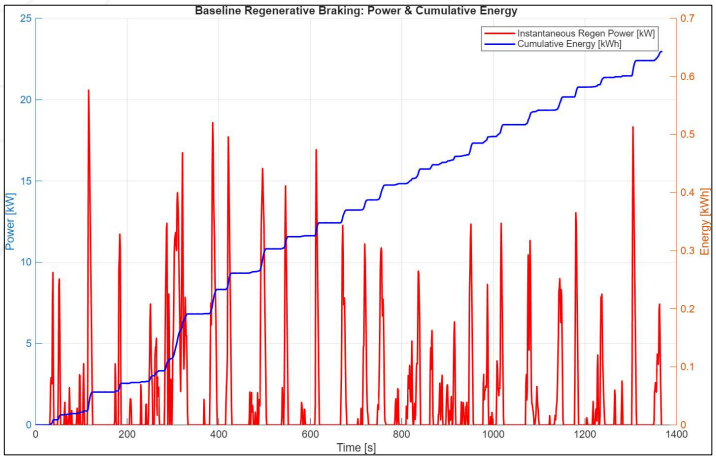
We start with the UDDS velocity profile, which is pre-recorded, experimentally measured data for an urban driving cycle that we can assume has minimal noise, and we know is evenly spaced.
To compute regenerative power, vehicle acceleration must also be calculated, since power is proportional to force times velocity. Acceleration is obtained numerically from the velocity using our custom betterDifferentiator().

The shown equation used for regenerative power is used given your vehicle parameters, and is evaluated with the test data for the baseline shown next.
The UDDS is a safe dataset to assume adequate conditions for our methods.
The figure overlays velocity (solid blue) and acceleration (dashed red) on the same axes and shows the relationship between speed changes and acceleration peaks that drive regenerative energy, useful for our computation.

Note that behavior of the differentiated acceleration curve - it highlights the discussed significance that error can have and the need for trust in the UDDS

dataset .

Results



Total Regen Energy: 0.6424 kWh

- From integration of previously shown regen power equation

Total Distance Traveled: 11.991 km

- From integration of UDDS given velocity data

These provide the baseline values necessary for the next analysis for comparison of noisy masked data.

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The figure on this slide shows instantaneous regenerative power over time, overlaid with the cumulative energy recovered. Integrating the power to give the Energy value, and doing this stepwise to see the accumulation of such, gives us our results.
We obtain the baseline value:
Total Regenerative Energy: 0.6424 kWh

Further, also needed for comparison, is: Total Distance Traveled: 11.991 km, arrived from by integrating the given UDDS velocity data.
These values provide the baseline reference for the next stage of analysis, where we will compare the effects of noise, quantization, and dropout on the same metrics.
You can now see how much energy is theoretically recoverable and the total distance covered under ideal conditions.

These results establish the reference point against which any sensor or data quality limitations can be evaluated.

Noise Analysis of Wheel Speed Data

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Methods and Assumptions

Need to model 'real' noisy data to determine sensor specifications.

- With UDDS dataset as baseline true values, apply:
  - Gaussian jitter v_jitter →
  - Quantization v_quant →
  - Dropout m_i (binary mask) →

$$v_{jitter}(t) = v_{true}(t) + n(t), \text{ where } n(t) \sim N(0, \sigma^2)$$

$$v_{quant}(t) = q \cdot \text{round}\left(\frac{v_{jitter}(t)}{q}\right)$$

$$v_{\text{meas}}(t_i) = m_i \cdot q \cdot \text{round}\left(\frac{v_{\text{true}}(t_i) + n_i}{q}\right)$$

- Existing MATLAB WheelNoiseApp is used to implement the above
  - All forms of noise present in a real life sensor accounted for
  - Noise affects derivates and integrals → determine a sensor constraint to maintain method usability

Apply this method by sweeping Gaussian σ to determine max allowable jitter.

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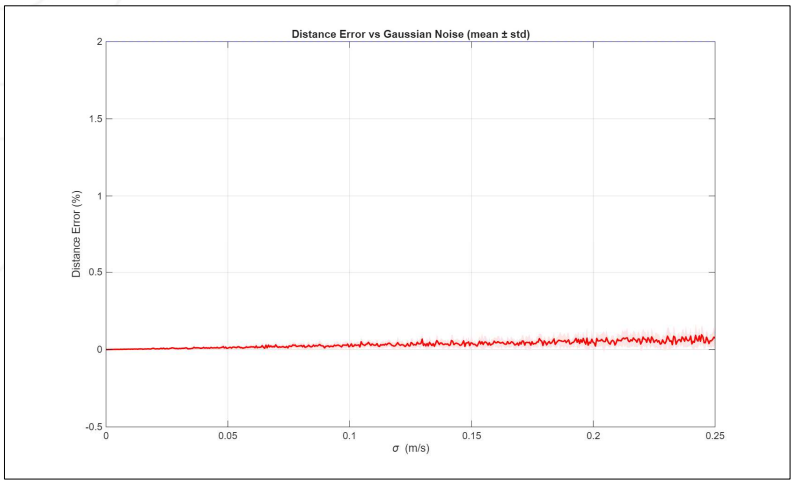
The UDDS velocity profile provides the "true" baseline signal.
Real sensors introduce three primary imperfections:
Jitter (Gaussian noise) - high-frequency random variation
Quantization - discretization errors from digital encoders
Dropout - random sample loss due to communication or timing glitches
The combined model is shown.
For our method to be valid, we assume that
Dropout and quantization are moderate; local spacing irregularities are small
Noise sources are independent of the true signal.

Therefore, the primary limiting dependent factor is the sensor. This will by far provide the greatest variability, and we know that our methods can handle uneven intervals. Data that is less smooth is more unknown. Therefore, we primarily analyze this factor.

We will sweep over gaussian sigma values to determine what the maximum deviation is for the random jitter that might arise. This will then become the sensor package's constraint value, as we will only be able to guarantee method Regen data approximation to that point.

This allows us to quantify the effect of sensor imperfection.

Results - Distance



Distance Error vs. Gaussian Nose is stable and not the limiting factor

- Distance error is stable over the analyzed range
- Integrated noise has little effects
- **Max σ allowed: >.25 m/s**

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This slide shows the percent error in total distance traveled as a function of Gaussian sensor noise (σ).
Thex-axis represents the noise magnitude applied to the velocity signal, and the y-axis shows the resulting percent error compared to the baseline.

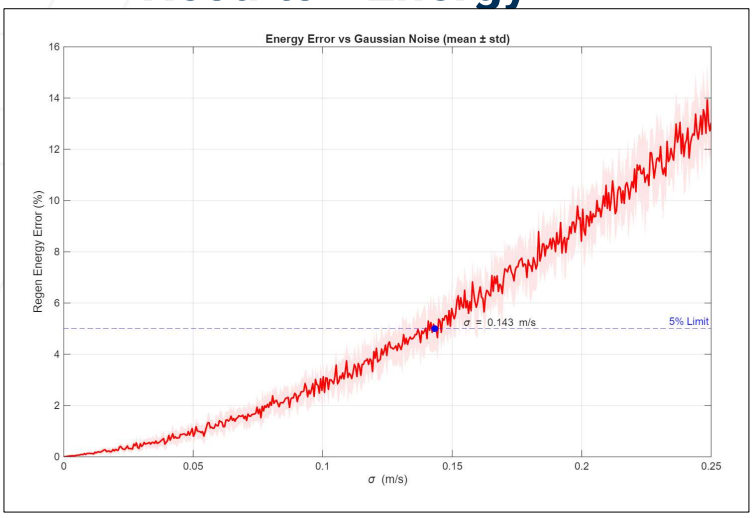
Notice that the distance error remains very small and nearly flat across the σ range we tested, up to our maximum of 0.25 m/s.
This behavior is expected because distance is obtained by integrating velocity, which naturally smooths out high-frequency noise.

Even with moderate jitter, the cumulative distance remains extremely close to the baseline, meaning our numerical integration is robust to realistic sensor noise.
This provides an important early insight: sensor jitter is less critical for total distance calculation, which only relies on integration of our data, but as we will see in the next slide, energy recovery is more sensitive due to differentiation steps.

The flatness of this curve also gives confidence in the stability and reliability of

our distance results.

Results - Energy



- Max σ allowed (distance 2%): 0.250 m/s
- Max σ allowed (energy 5%): 0.145 m/s

Overall allowable σ: 0.145 m/s

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This plot shows percent error in total regenerative energy as a function of Gaussian sensor noise applied to the UDDS velocity signal.

Energy calculation is sensitive because it involves both velocity and acceleration (derivative of velocity), which amplifies high-frequency noise.

The solid line is the mean error across multiple noise realizations, while the shaded region represents ±1 standard deviation, showing the natural variability due to the random noise.

The +/- term was computed by repeating the simulation multiple times for each sigma, then taking the standard deviation across realizations. This gives a statistical measure of how much the error could fluctuate in practice.

Major observations:

Maximum allowable σ to keep energy error ≤5% is 0.145 m/s, lower than distance because derivatives amplify noise.
The standard deviation +/-0.03 m/s provides a measure of confidence: the error is likely to remain within this range even with the stochastic nature of the noise.This is not a measure to use as a constraint, but as a validity metric.

For our energy calculations to remain within 5% error, wheel-speed sensor jitter should not exceed 0.145 m/s.

This will retain confidence in the numerical methods we developed for you.

- High accuracy custom methods give baseline:
  - Total Regenerative Energy: 0.06424 kWh
  - Total Distance Traveled: 11.991 km
- Noise analysis determines **maximum sensor jitter of 0.145 m/s**

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In this analysis, we developed high-accuracy differentiation and integration tools (betterDifferentiator and betterIntegrator) to process UDDS data according to the specified parameters, establishing a baseline with a total regenerative energy of 0.06424 kWh over 11.991 km. This confirmed the drivetrain's performance under ideal measurement conditions. We then performed a noise analysis for the sensor package, incorporating Gaussian jitter, dropout, and quantization effects, and compared the resulting errors to determine that the maximum allowable sensor jitter is 0.145 m/s. This threshold provides a clear quantitative limit for sensor noise, ensuring that regenerative braking and distance calculations remain accurate. Sensors sensors must meet or exceed this jitter specification to maintain reliable energy and distance measurements, and reliance on our numerical methods.

Thank you for your business!